

AGHA ABRAR KHAN

Full Stack AI Engineer | Python & TypeScript | Generative AI & RAG

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PROFESSIONAL SUMMARY

Full Stack ML Engineer with 4+ years of freelance and professional software delivery experience, combining full-stack development with hands-on work in machine learning, generative AI, and intelligent automation. Skilled in building responsive, API-driven web applications using Python, FastAPI, JavaScript, TypeScript, React, Next.js, Node.js, and SQL/NoSQL databases. Experienced in integrating AI models into applications, developing backend APIs, and connecting data workflows with interactive user interfaces. Built RAG systems, LLM-powered assistants, and computer-vision applications using OpenAI APIs, Hugging Face, LangChain, embeddings, and vector search. Strong problem solver with experience delivering projects from requirements and architecture through integration, testing, deployment, and documentation.

TECHNICAL SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Tailwind CSS, Responsive UI, Component Architecture, Dashboard Development, API-Driven Interfaces, State Management

Backend & APIs: Node.js, Express.js, Python, FastAPI, Flask, REST APIs, Webhooks, Microservices, API Design, API Integration, Server-Sent Events, WebSockets

Databases & ORMs: MySQL, Prisma, PostgreSQL, Supabase, MongoDB, SQL, Relational Schema Design, Data Modeling, Query Optimization Basics

Modern TypeScript Stack: TypeScript, Tailwind CSS, Prisma, MySQL, tRPC Familiarity, Type-Safe API Patterns, Full-Stack Product Architecture, Reusable Code Practices

Security & Fintech Basics: Secure API Practices, Authentication Concepts, Authorization/RBAC, Input Validation, Data Integrity, Sensitive Data Handling, Audit Logging Basics

AI & Automation: OpenAI API, Hugging Face, Ollama, LangChain, LangGraph, RAG, AI Chatbots, AI Copilots, Prompt Engineering, Custom Backend Automation

Cloud, DevOps & Tools: Docker, Git, GitHub, Vercel, Supabase, AWS/S3 Basics, CI/CD Familiarity, Environment Configuration, Deployment Troubleshooting

Collaboration: Requirements Understanding, Solution Architecture, Product Collaboration, Documentation, Agile Delivery, Code Reviews, Stakeholder Communication

EDUCATION

Sukkur IBA University
BS Computer Science (BSCS)

Sukkur, Pakistan

PROFESSIONAL EXPERIENCE

Freelancing

2021 – Present

Freelance Full Stack & AI Developer

Remote

- Developed responsive full-stack web applications using TypeScript, JavaScript, React, Next.js, Tailwind CSS, Node.js, Express.js, Python, FastAPI, and Flask.
- Built reusable frontend components, dashboards, responsive user interfaces, API-driven interfaces, and state-managed application workflows.
- Designed and integrated REST APIs, webhooks, authentication systems, role-based access control, input validation, and real-time functionality using Server-Sent Events and WebSockets.
- Worked with PostgreSQL, MySQL, MongoDB, Supabase, Prisma, and SQL for relational schema design, data modeling, database integration, and query handling.
- Integrated AI capabilities into web applications using OpenAI API, Hugging Face, Ollama, LangChain, LangGraph, RAG pipelines, AI chatbots, AI copilots, and prompt engineering.
- Implemented backend automation workflows and integrations with third-party APIs to support application-specific business processes and AI-powered features.
- Applied secure development practices including authentication, authorization, sensitive data handling, data integrity, audit logging concepts, and secure API design.
- Used Docker, Git, GitHub, Vercel, Supabase, and cloud deployment tools for application development, version control, environment configuration, deployment, and troubleshooting.
- Handled requirements understanding, solution planning, development, testing, documentation, deployment, maintenance, and stakeholder communication throughout the development lifecycle.

KEY PROJECTS

- DevMind AI – Multi-Agent Code Intelligence Workspace:** Built a full-stack AI workspace using Next.js, TypeScript, FastAPI, PostgreSQL, pgvector, Redis, Celery, and LangGraph to index GitHub repositories and technical documents and answer codebase questions using multi-agent Retrieval-Augmented Generation. Implemented hybrid vector and full-text retrieval, agent-based code analysis, reviewer-validated responses, file-level citations, SSE streaming, JWT authentication, background ingestion, and Docker-based deployment.
- MCP-Powered GitHub Repository Assistant:** Built a Next.js and TypeScript application integrating OpenAI and GitHub REST APIs through an MCP-style tool layer to analyze repository activity, prioritize issues, summarize pull requests, identify documentation gaps, and generate sprint plans.

- **AI-Powered Clinical Decision Support System:** Engineered a large-scale AI-enabled clinical decision support platform using TypeScript, NestJS, FastAPI, Prisma, and PostgreSQL, featuring patient management, role-based access control, audit logging, deterministic medication-safety checks, secure clinical data workflows, and an LLM gateway with de-identification and structured response validation.
- **Resume Ranking System:** Built a Python and Flask application that extracts text from uploaded PDF resumes, preprocesses content, and ranks resumes against job descriptions using TF-IDF vectorization and cosine similarity, presenting results through a web dashboard.
- **Transparent Auction House – Final Year Project:** Developed a React and Flask auction application with JWT-protected bidding APIs and bid-history tracking. Implemented bidding-behavior feature extraction and a model-inference module for detecting suspected shill bidding.

CERTIFICATIONS

- **Claude with Amazon Bedrock** – Anthropic Education
- **MCP: Advanced Topics** – Anthropic Academy
- **Python and Flask Framework Complete Course** – Udemy
- **Generative AI and Artificial Intelligence (AI) for Leaders** – Udemy
- **Complete JavaScript with HTML5, CSS3 from Zero to Expert – 2026** – Udemy
- **NodeJS Masterclass (Express, MongoDB, OpenAI)** – Udemy
- **Flutter Masterclass (Dart, API & More)** – Udemy
- **Diploma in Information Technology (DIT)** – Trade Testing Board Sindh
- **Certificate in Information Technology (CIT)** – Prime Institute of Science & Technology, Sukkur
- **English Language Diploma** – Prime Institute of Science & Technology and English Language, Sukkur

PORTFOLIO



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LANGUAGES

English, Urdu, Sindhi